

Adaboost { Decision tree }

f_1 f_2 f_3 f_4 O/p Weight

- - - - Yes $1/7$

- - - - No $1/7$

- - - - $1/7$

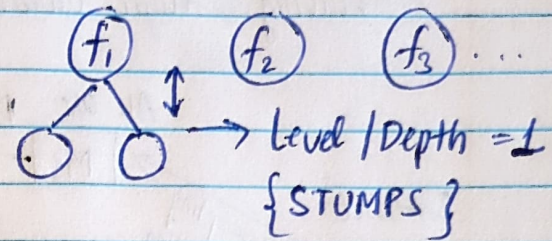
- - - - $1/7$ → wrongly predicted by model

- - - - $1/7$

- - - - $1/7$

- - - - $1/7$

Σ = 1



• Weak learner

① Total Error = 1 record (weight) = $1/7$

② Performance of Stump (1 level Decision tree)

$$P_s = \frac{1}{2} \log_e \left(\frac{1 - TE}{TE} \right)$$

$$= \frac{1}{2} \log_e \left(\frac{1 - 1/7}{1/7} \right)$$

$$= 0.895$$

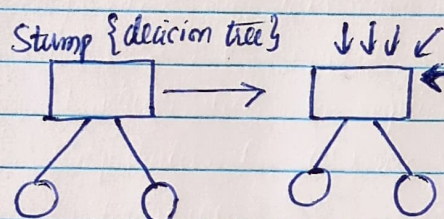
③ Updates the weights of all records

New Sample Weight

$$\begin{aligned} \text{Correct record} &= \text{old weight} * e^{-P_s} = \frac{1}{7} * e^{-0.895} = 0.05 \\ \text{Incorrect record} &= \text{old weight} * e^{P_s} = \frac{1}{7} * e^{0.895} = 0.349 \end{aligned}$$

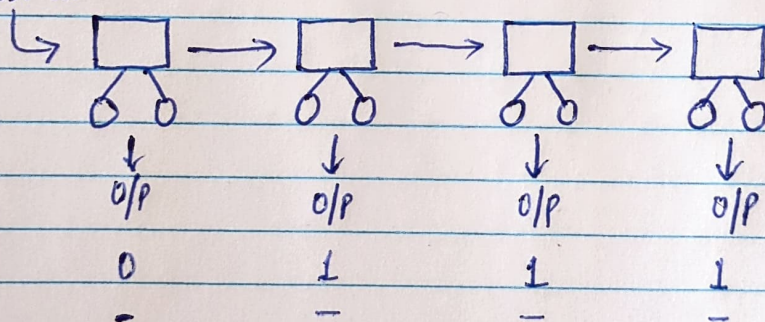
<u>new weight</u>		<u>Normalized weight</u>	<u>Buckets</u>
0.05		$0.05 \div 0.649 = 0.07$	$[0 - 0.07]$
0.05		" = 0.07	$[0.07 - 0.14]$
0.05		= 0.07	$[0.14 - 0.21]$
0.349	→ this total	$0.349 \div 0.649 = 0.53$	$[0.21 - 0.747]$
0.05	probability is	= 0.07	$[0.747 - 0.751]$
0.05	not 1. So we	= 0.07	"
0.05	do normalization	= 0.07	"
$\Sigma = 0.649$			

This record is having higher bucket size



• Since the bucket size is higher it gets selected mostly and go to the second tree

Test data



{classifier} → majority → 1
{regressor} → average

The wrong records from 1st stump goes to 2nd stump

→ The wrong records from 2nd stump goes to 3rd stump

→ Similarly for other stumps....